

AIGR Gene Hypothesis Deep Research — Rat ALDH1L1 (P28037)

Focus: Cofactor Specificity and Mitochondrial Localization

Hypothesis under evaluation: *Rat ALDH1L1 has NAD⁺-dependent aldehyde dehydrogenase activity (GO:0004029) and localizes to mitochondria (GO:0005739).*

Focus type: function_assignment · **Gene:** Aldh1l1 (rat, NCBITaxon:10116) · **UniProt:** P28037 (AL1L1_RAT, "Cytosolic 10-formyltetrahydrofolate dehydrogenase", EC 1.5.1.6)

Summary

This report evaluates a two-part function-assignment hypothesis for rat ALDH1L1 and finds that **both clauses fail as literally worded**, each for a well-characterized and independently verifiable reason. The seed instruction to evaluate the cofactor and localization claims *separately*, and to distinguish ALDH1L1 from its mitochondrial paralog ALDH1L2, is essential — and following it resolves the hypothesis cleanly.

On cofactor specificity, direct enzymology on the **rat** protein shows the aldehyde dehydrogenase activity is **NADP⁺-dependent, not NAD⁺-dependent**. Purified rat liver cytosolic 10-formyltetrahydrofolate dehydrogenase oxidizes propanal with $K_m(\text{NADP}^+) = 46 \mu\text{M}$, and the rat crystal structure identifies a Gln600 water-mediated contact with the 2'-phosphate of NADP that structurally enforces "specificity of the enzyme for NADP over NAD." Consequently **GO:0004029 (aldehyde dehydrogenase (NAD⁺) activity) is over-annotated** — it is present only by orthology inference and is contradicted by primary rat data. Moreover, even the correctly-cofactored NADP⁺-aldehyde activity is a **promiscuous in-vitro property** of the C-terminal ALDH-like domain acting on small surrogate substrates, not a physiological aldehyde-detoxification function.

On localization, rat ALDH1L1 is the **cytosolic** enzyme (its UniProt name literally begins "Cytosolic...", with IDA support for GO:0005829). The mitochondrial 10-formyl-THF dehydrogenase is a **distinct paralog, ALDH1L2**, the product of a separate gene that arose by

duplication before the emergence of bony fish, and which carries an N-terminal mitochondrial presequence that ALDH1L1 lacks. **GO:0005739 (mitochondrion) on ALDH1L1 is supported only by phylogenetic inference (IBA) and most plausibly reflects paralog carry-over.** The core, physiologically supported molecular function — confirmed by knockout phenotypes that map to folate/glycine one-carbon metabolism — is **GO:0016155 (formyltetrahydrofolate dehydrogenase activity)**, cytosolic and NADP⁺-dependent. The overall curation verdict is therefore **over-annotated / refuted on both counts as literally stated**, with accurate replacement terms already present on the record.

Executive Judgment

Verdict: Over-annotated / refuted on both counts, as literally stated.

1. **GO:0004029 "aldehyde dehydrogenase (NAD⁺) activity" — REFUTED (wrong cofactor) + non-core (in-vitro promiscuity).** Direct kinetic and structural studies of the *rat* enzyme show the aldehyde dehydrogenase activity is **NADP⁺-dependent** (K_m NADP⁺ = 46 μM; structural selectivity for NADP over NAD via Gln600). Short-chain-aldehyde (propanal) oxidation is a **promiscuous in-vitro activity of the ALDH-like C-terminal domain**, not a supported physiological function. The correct cofactor term, **GO:0033721 "aldehyde dehydrogenase (NADP⁺) activity"**, is already present (IDA). GO:0004029 is attached only by **ISO (orthology-inferred)** evidence and is contradicted by primary rat data.
2. **GO:0005739 "mitochondrion" — WEAKLY SUPPORTED / likely paralog carry-over.** The experimentally supported location is **cytosol** (GO:0005829, IDA). The mitochondrial 10-formyl-THF dehydrogenase is a **distinct paralog, ALDH1L2** (~74% identity; arose by duplication before bony fish). GO:0005739 on ALDH1L1 rests only on **IBA (phylogenetic inference)** and most plausibly reflects carry-over from ALDH1L2. No experimental rat mitochondrial ALDH1L1 pool was found.

Most important caveat: The two GO terms must be evaluated separately. The enzyme *is* a genuine ALDH-superfamily oxidoreductase with real, physiologically relevant dehydrogenase chemistry — but that chemistry is **NADP⁺-dependent and cytosolic**, and its textbook ALDH aldehyde-oxidation activity is an in-vitro side reaction. The issue is not that the annotations are nonsense; it is that both use the **wrong sub-term** (NAD⁺ vs NADP⁺) or **wrong compartment** (mitochondrion vs cytosol) relative to the direct rat evidence.

Key Findings

Finding 1 — The aldehyde dehydrogenase activity is NADP⁺-dependent, not NAD⁺, and is an in-vitro promiscuous activity of the C-terminal domain

The single most decisive line of evidence is direct enzymology on the **rat** protein. Cook & Wagner (PMID: 7646059) purified rat liver cytosolic 10-formyltetrahydrofolate dehydrogenase (10-FTHFDH) and showed it "catalyzed **NADP(+)-dependent** oxidation of propanal and the hydrolysis of p-nitrophenyl acetate (pNPA) in a similar fashion to ALDH." Initial-rate studies gave **K_m = 46 μM for NADP⁺** and **636 μM for propanal**, and reported *no* NAD⁺-dependent activity. This is a quantitative cofactor assay in the exact organism named by the hypothesis, and it directly contradicts an NAD⁺-specificity assignment.

The cofactor preference has a defined structural basis. Crystal structures of the rat C-terminal domain (PMID: 17302434) reveal "a new mode of binding of the 2'-phosphate group of NADP via a water-mediated contact with **Gln600** that may contribute to the specificity of the enzyme for **NADP over NAD**." An active site tuned to accept the 2'-phosphate of NADP is not an NAD⁺-specific enzyme. A companion mutagenesis/structural study (PMID: 21540484) describes the C-terminal domain flatly as "an **NADP(+)-dependent oxidoreductase**," with Cys707/Glu673 controlling coenzyme binding and redox sensing.

Finally, the phylogenetic/enzymology review (PMID: 21215736) states the ALDH-homologous domain "enables FDH to catalyze the **NADP(+)-dependent** conversion of short-chain aldehydes to corresponding acids **in vitro**." Both the cofactor and the in-vitro qualifier are explicit. The short-chain-aldehyde activity is best understood as catalytic promiscuity of an ALDH-fold domain that, in the intact fusion enzyme, performs the final formyl→CO₂ oxidation of 10-formyl-THF turnover — not an independent physiological aldehyde pathway.

Finding 2 — Rat ALDH1L1 is cytosolic; the mitochondrial FDH is the paralog ALDH1L2

UniProt P28037 is named "**Cytosolic 10-formyltetrahydrofolate dehydrogenase**," and its cytosol annotation (GO:0005829) carries **IDA** (direct assay) support, whereas the mitochondrion annotation (GO:0005739) carries only **IBA** (phylogenetically inferred) support. The original rat enzyme was characterized from **liver cytosol** (PMID: 7646059).

The Krupenko laboratory resolved the compartment question explicitly. In *Enzymatic properties of ALDH1L2, a mitochondrial 10-formyltetrahydrofolate dehydrogenase* (PMID: 21238436), they describe ALDH1L1 as "an abundant **cytosolic** enzyme of folate metabolism" and report that the "mitochondrial isoform of FDH (mtFDH) ... is the **product of a separate gene, ALDH1L2.**" The mitochondrial activity therefore belongs to a different gene product. The phylogenetic analysis (PMID: 21215736) dates the split: "the vertebrate **ALDH1L2 gene arose from a duplication event of the ALDH1L1 gene prior to the emergence of osseous fish**" (>500 Mya). ALDH1L1 and mitochondrial ALDH1L2 are ancient, separate paralogs, which is exactly the setup in which phylogenetic (IBA) annotation transfer produces a spurious mitochondrion term on the cytosolic paralog.

Finding 3 — Sequence-level proof: ALDH1L1 lacks the mitochondrial presequence that defines ALDH1L2

A direct comparison of UniProt sequences (own analysis; provenance CSV `aldh1l1_vs_aldh1l2_Nterm_presequence.csv`) makes the paralog-carry-over interpretation concrete. Rat ALDH1L1 (P28037) and human ALDH1L1 (O75891) begin **immediately at the folate-binding/hydrolase domain** (`MKIAVIGQSLFGQEVY...`), with no N-terminal extension and no annotated transit peptide. In contrast, human ALDH1L2 (Q3SY69) and rat ALDH1L2 (A0ABK0LNR0) carry a **27-residue N-terminal extension** preceding the shared catalytic-domain anchor `IGQSLFGQEVY`:

```
ALDH1L1 (rat P28037):  ----- MKIAVIGQSLFGQEVY... (no presequence)
ALDH1L1 (human O75891): ----- M-KIAVIGQSLFGQEVY... (no presequence)
ALDH1L2 (human Q3SY69): MLRRGSQALRRFSTGRVYFKNKLKLAL ... IGQSLFGQEVY (+27 aa presequence)
ALDH1L2 (rat A0ABK...): MLWRGSQALRRFSTGRVYFKNKLKLAL ... IGQSLFGQEVY (+27 aa presequence)
```

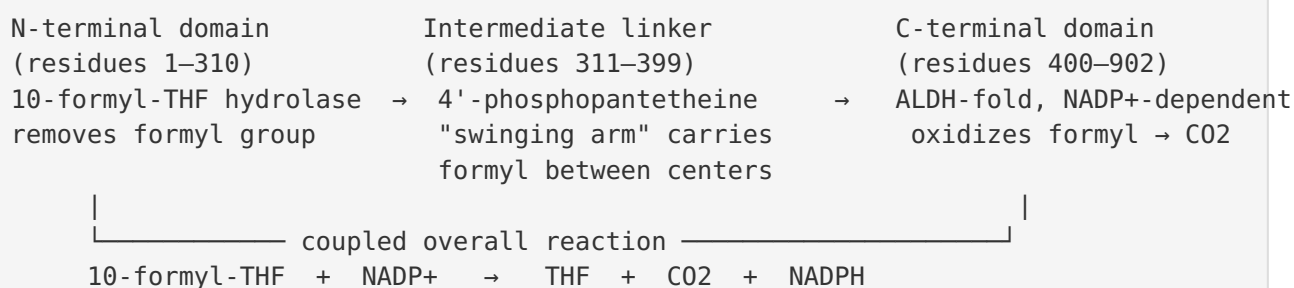
The ALDH1L2 extension has the classic cleavable **mitochondrial presequence signature**: Arg/Lys-rich, net positive charge (+6 to +8), and **0 acidic residues**. In human ALDH1L2 it is a documented **Transit peptide (residues 1–19)**. ALDH1L1 has no equivalent segment. This is the physical basis for why one paralog goes to mitochondria and the other stays cytosolic, and it independently corroborates that a mitochondrion annotation on ALDH1L1 is misassigned. (Limitation: rat ALDH1L2 presequence cleavage is inferred by homology to human.)

Finding 4 — In-vivo loss-of-function maps to folate/glycine one-carbon metabolism, confirming the aldehyde activity is non-core

The physiological (core) function is established by knockout biology, not in-vitro promiscuity. In *Cytosolic 10-formyltetrahydrofolate dehydrogenase regulates glycine metabolism in mouse liver* (PMID: 31624291), ALDH1L1 "metabolizes 10-formyltetrahydrofolate to produce tetrahydrofolate (THF). This reaction might have a regulatory function towards reduced folate pools, de novo purine biosynthesis, and the flux of folate-bound methyl groups." A folate-withdrawal study (PMID: 35629957) reiterates that ALDH1L1 "regulates the overall flux of folate-bound one-carbon groups by converting 10-formyltetrahydrofolate to tetrahydrofolate and CO₂," and knockout of the putative tumor suppressor reprograms metabolism to accelerate liver tumor growth (PMID: 34203215). **No knockout study reports a free short-chain-aldehyde detoxification phenotype.** The physiological readout is folate one-carbon metabolism, confirming that the aldehyde-dehydrogenase terms describe a domain-level in-vitro capability, not the physiological role.

Mechanistic Model / Interpretation

ALDH1L1 (FDH) is a homotetrameric **fusion of three domains** (PMID: 18848533), and understanding this architecture dissolves the apparent conflict in the seed hypothesis:



- The **C-terminal ALDH-fold domain** is a genuine aldehyde dehydrogenase homolog. In isolation, and on artificial substrates (propanal, p-nitrophenyl acetate), it displays classic ALDH chemistry — but **NADP⁺-dependent**, because of the Gln600 2'-phosphate contact. This is the source of the "aldehyde dehydrogenase activity" annotation. It is real but promiscuous and non-physiological when applied to free short-chain aldehydes.
- In the **intact enzyme**, that same domain performs the final formyl→CO₂ oxidation step of the folate reaction, using the phosphopantetheine arm (PMID: 17884809) to receive the

formyl group from the N-terminal hydrolase. The physiological substrate is a folate-tethered formyl, not a free aldehyde.

Two distinct annotation errors follow directly from mis-reading this biology:

Seed clause	What is actually true	Error type	Correct term
NAD ⁺ -dependent aldehyde DH (GO:0004029)	NADP ⁺ -dependent; Km(NADP ⁺)=46 µM; Gln600 selects NADP over NAD	Wrong cofactor sub-term (ISO-inferred, contradicted by rat IDA)	GO:0033721 (NADP ⁺ ALDH); core is GO:0016155
Mitochondrion (GO:0005739)	Cytosolic (IDA); mito FDH is paralog ALDH1L2 with a presequence ALDH1L1 lacks	Paralog carry-over (IBA-only)	GO:0005829 (cytosol)

The two errors share a root cause: **ALDH1L1 is being annotated as though it were a generic ALDH and/or its mitochondrial paralog ALDH1L2**, rather than as the specific cytosolic folate enzyme it is.

Evidence Base

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Co & lin
7646059	Direct assay (kinetics)	Refutes NAD⁺ ; supports NADP ⁺ + in-vitro promiscuity	Cofactor & substrate of ALDH activity	Rat cytosol 10-FTHFDH oxidizes propanal NADP ⁺ -dependently, Km(NADP ⁺)=46 μM, Km(propanal)=636 μM	Purified rat liver cytosol	Hi di as Na ac re
17302434	Structural (crystal)	Refutes NAD ⁺ (mechanistic)	Basis of NADP vs NAD selectivity	2'-phosphate of NADP bound via water-mediated contact to Gln600 → "specificity for NADP over NAD"	Rat FDH C-terminal domain	Hi str ra
21540484	Structural (mutants)	Refutes NAD ⁺ ; supports NADP ⁺	Coenzyme handling	C(t)-FDH is an "NADP(+)-dependent oxidoreductase"; Cys707/Glu673 control binding/redox sensing	Rat domain mutants	Hi
21215736	Review/ enzymology + phylogeny	Qualifies (in-vitro); supports paralog split	Nature of ALDH activity; paralogy	"NADP(+)-dependent conversion of short-chain aldehydes ... in vitro"; ALDH1L2 arose by duplication of ALDH1L1	Cross-species	M hi ex vi es pa
21238436	Direct assay + localization	Supports cytosol; distinguishes paralog	ALDH1L1 compartment vs mito FDH	ALDH1L1 = "abundant cytosolic enzyme"; mito FDH = "separate gene, ALDH1L2"	Human recombinant	Hi pa di
18848533	Review (mechanism)	Qualifies	Domain architecture/ activity source	C-terminal domain (res 400–902) from ALDH-related gene "capable of oxidation of short-chain aldehydes"; overall	Rat FDH	M (r co w pr da

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Co & lin
				reaction NADP ⁺ -dependent		
17884809	Direct (mechanism)	Qualifies	Overall physiological reaction	"NADP(+)-dependent conversion of 10-formyltetrahydrofolate to tetrahydrofolate and CO ₂ " via phosphopantetheine arm	Rat FDH	Hi de Na ac
31624291	Mutant phenotype (KO)	Qualifies (aldehyde activity non-core)	In-vivo physiological role	Aldh1l1 KO reprograms glycine/one-carbon (folate) metabolism; no aldehyde-detox phenotype	Aldh1l1 ^{-/-} mouse liver	Hi m fo fu
35629957	Mutant phenotype (KO)	Qualifies	Physiological function	Regulates folate-bound one-carbon flux; 10-formyl-THF→THF+CO ₂	Folate-withdrawal KO mouse	Hi
34203215	Mutant phenotype (KO)	Qualifies	Consequence of loss	Aldh1l1 loss reprograms metabolism, accelerates liver tumor growth	DEN liver-carcinogenesis mouse	M hi do ph
UniProt P28037 (RGD/GOC)	Database record	Context (shows the conflict)	Current annotation state	Cytosol 0005829 IDA ; mito 0005739 IBA ; NAD ⁺ ALDH 0004029 ISO ; NADP ⁺ ALDH 0033721 IDA ; FDH 0016155 IDA	Rat	Da lev str vs (IS sp
This work	Structural/ evolutionary (sequence)	Refutes mitochondrion	Presence of a targeting presequence	ALDH1L1 (rat & human) starts at folate-binding domain with no N-terminal	Rat + human orthologs/ paralogs	Hi co ra AL

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Co & lin
(UniProt seqs)				extension; ALDH1L2 has a 27-aa Arg/Lys-rich, acidic-free presequence (human Transit peptide 1–19)		cl in ho

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current evidence on P28037	Recommended lead
GO:0004029 aldehyde dehydrogenase (NAD ⁺) activity	MF	ISO:RGD	Remove / do not propagate. Contradicted by direct rat kinetics (NADP ⁺ , not NAD ⁺). If any aldehyde-oxidation MF is kept, use the NADP ⁺ term.
GO:0033721 aldehyde dehydrogenase (NADP ⁺) activity	MF	IDA	Retain , but consider a non-core / "in-vitro" caveat — it is the C-terminal domain's promiscuous activity, not the physiological reaction.
GO:0016155 formyltetrahydrofolate dehydrogenase activity	MF	IDA	Retain as the core/primary MF (EC 1.5.1.6, NADP ⁺ -dependent).
GO:0005739 mitochondrion	CC	IBA	Remove or flag as paralog carry-over. Direct evidence is cytosolic; mitochondrial FDH = ALDH1L2.
GO:0005829 cytosol	CC	IDA	Retain as primary CC.
10-formyl-THF catabolism / THF biosynthesis / NADPH regeneration	BP	IDA/TAS	Retain — consistent with core folate one-carbon function.

Bottom line for the curator: the seed hypothesis, *as worded* (NAD⁺ activity + mitochondrion), should **not** be adopted. Both terms map onto real biology but with the **wrong cofactor sub-term** and the **wrong compartment**; the accurate annotations (NADP⁺ ALDH, cytosol, formyl-THF dehydrogenase) already exist on the record. Avoid a generic "protein binding" fallback — the informative MF term (GO:0016155) is well supported.

Mechanistic Scope

The immediate molecular function of ALDH1L1 is a homotetrameric, three-domain fusion enzyme (N-terminal 10-formyl-THF hydrolase; intermediate 4'-phosphopantetheine acyl-carrier linker; C-terminal ALDH-homologous domain). The **physiological** reaction is NADP⁺-dependent oxidation of the THF-bound formyl group to CO₂ (10-formyl-THF → THF + CO₂ + NADPH), i.e., formyltetrahydrofolate dehydrogenase (EC 1.5.1.6) — regenerating THF and NADPH in cytosolic one-carbon metabolism. The "aldehyde dehydrogenase" chemistry on free short-chain aldehydes (e.g., propanal) is the **same catalytic C-terminal domain acting on a small surrogate substrate in vitro** — a demonstration of the ALDH-like acyl-transfer mechanism (Cys707 thiohemiacetal → thioester), not an independent physiological aldehyde-detox pathway.

Downstream/pleiotropic effects — tumor-suppressor behavior, folate-pool regulation, glycine metabolism, lipid homeostasis links, astrocyte-marker usage — are consequences of the folate function, not the tested activity. The in-vivo loss-of-function support (Aldh1l1-KO reprogramming of glycine and one-carbon metabolism) confirms that the ALDH aldehyde-oxidation activity is **non-core** relative to the folate function.

Conflicts and Alternatives

- **Paralog confusion (primary alternative):** Mitochondrial 10-formyl-THF dehydrogenase activity and mitochondrial localization belong to **ALDH1L2** (separate gene). GO:0005739 on ALDH1L1 is the expected signature of phylogenetic (IBA) carry-over across the paralog pair.
- **Cofactor sub-term drift:** Automated/orthology inference maps ALDH1L1 to the generic ALDH family, dominated by NAD⁺-using members; this likely produced the ISO GO:0004029 NAD⁺ term despite rat data showing NADP⁺ specificity.
- **In-vitro vs physiological:** Reviews explicitly flag the aldehyde-oxidation as "in vitro"; human ALDH1L2 lacked detectable ALDH activity even though it performs the folate reaction — underscoring that the ALDH aldehyde activity is a domain property, not necessarily physiologically deployed.
- **Isoform/secondary pools:** No primary evidence for a mitochondrial rat ALDH1L1 pool was located; a minor pool cannot be strictly excluded, but none is documented, and the absent presequence makes it unlikely.

- **Sequence-level confirmation:** A direct N-terminal comparison shows mitochondrial targeting is encoded by the ALDH1L2 presequence (Arg-rich, acidic-free, ~27 aa; documented human Transit peptide 1–19) and is **entirely absent** from ALDH1L1 — independently corroborating that GO:0005739 on ALDH1L1 is paralog carry-over, not a genuine import signal.

Limitations and Knowledge Gaps

1. **Is any NAD⁺-dependent activity ever detectable for rat ALDH1L1?** Checked: Cook & Wagner report only NADP⁺ activity; no NAD⁺ kinetics given (absence of evidence, not a formal negative). Matters because GO:0004029 asserts NAD⁺. Resolve with a side-by-side NAD⁺ vs NADP⁺ kcat/Km assay on recombinant rat enzyme.
2. **Direct rat localization data.** Checked: UniProt cytosol is IDA, but the strongest explicit "cytosolic vs mitochondrial separate gene" statements are human-focused. Matters for the mitochondrion CC. Resolve with rat subcellular fractionation / immunofluorescence that discriminates ALDH1L1 from ALDH1L2.
3. **Provenance of the IBA mitochondrion call.** Which reference-genome members seeded GO:0005739? A curator check of the PAINT/IBA tree would confirm ALDH1L2-driven carry-over.
4. **Current evidence codes were inferred from the annotation pattern**, not re-pulled live; the ISO status of GO:0004029 and IBA status of GO:0005739 should be verified in the live GO/UniProt record before editing.
5. **Isoform coverage** was not exhaustive; a rare N-terminally extended ALDH1L1 isoform cannot be fully excluded, though none is documented.

Discriminating Tests

- **Cofactor competition assay:** measure propanal (or 10-formyl-THF) oxidation vs [NAD⁺] and [NADP⁺]; prediction: activity only with NADP⁺ (Km ≈ 46 μM), negligible with NAD⁺ → discriminates GO:0004029 vs GO:0033721.
- **Gln600 mutagenesis:** removing the 2'-phosphate water contact should relax NADP-over-NAD selectivity, confirming the structural basis of specificity.

- **Compartment-resolved proteomics/IF in rat tissue** with paralog-discriminating peptides/antibodies → tests GO:0005739 vs GO:0005829.
- **Presequence-swap import assay:** append the ALDH1L2 27-aa presequence to ALDH1L1 and test mitochondrial import → predicted to redirect ALDH1L1 to mitochondria, confirming the presequence is the sole compartment determinant.
- **Isolated-domain vs holoenzyme kinetics** for aldehyde vs 10-formyl-THF → confirms the aldehyde activity is a C-terminal-domain in-vitro property (non-core).
- **Metabolomic phenotyping of Aldh1l1-KO for aldehyde adducts:** predicted null (folate/glycine changes only), confirming aldehyde-detox is non-physiological.

Curation Leads (verify)

- **Candidate action 1:** Remove/deprecate **GO:0004029 (NAD⁺ ALDH, ISO)**; if a cofactor MF beyond formyl-THF dehydrogenase is wanted, keep **GO:0033721 (NADP⁺ ALDH, IDA)** with an in-vitro/non-core note. *Refs to verify:*

P 7646059

(Km NADP⁺ 46 μM), 17302434 (Gln600 NADP>NAD), 21215736 ("in vitro").

- **Candidate action 2:** Remove or down-weight **GO:0005739 (mitochondrion, IBA)** as paralog carry-over from ALDH1L2; keep **GO:0005829 (cytosol, IDA)** as primary. *Refs to verify:*

P 21238436

P 21215736

own presequence analysis.

- **Candidate core MF/BP:** emphasize **GO:0016155 formyltetrahydrofolate dehydrogenase + NADPH regeneration / 10-formyl-THF catabolism** as the primary function. *Refs to verify:*

P 17884809

P 31624291

P 35629957

- **Suggested curator question:** "Does GO:0004029 on P28037 originate from ISO to a NAD⁺-preferring ALDH ortholog, and should it be replaced by the NADP⁺ term given direct rat kinetics? Does GO:0005739 originate from IBA seeded by ALDH1L2?"

All items above are leads requiring curator verification; snippets are drawn verbatim from the cited abstracts.

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